

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think>
we know john drives for the first bit at 60 miles per hour for 3 hours so we do 60 times 3 which equals 180 so now we know he's 180 miles from home
then we know he then tries to get home in 4 hours but gets stuck in traffic for 2 hours so we know now he has 4 hours take away the 2 hours of traffic which leaves us with 2 hours to travel
then we know for the first part of this 2 hours travel he does 30 miles per hour for half an hour so we do 30 miles per hour times half an hour which equals 15 so now we know he's travelled another 15 miles which takes him from the original 180 miles down to 180 take away the 15 which equals 165
then we know for the rest of the time which is the last half an hour he's travelling at 80 miles per hour so we do 80 miles per hour times half an hour which equals 40
so now we know he's travelled another 40 miles in this last bit of the journey so we take the 165 miles he had left and take away the 40 miles he's travelled in this last part which equals 125
so now we know he's 125 miles away from home
</think><answer>125</answer>

